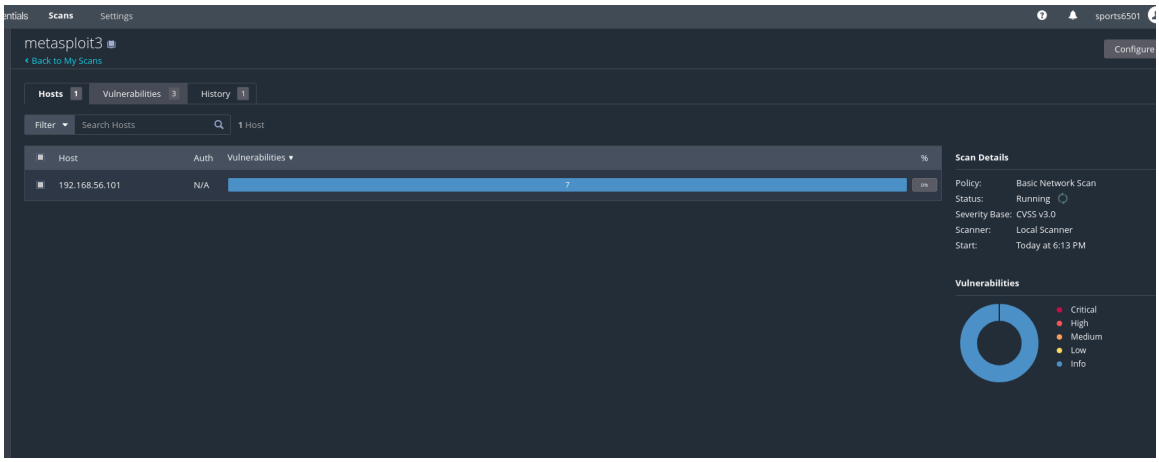
Comparative Analysis: Nessus Basic Network Scan vs Web Application Tests

Objective: Compare two Nessus scan templates run against the same target and analyze the differences in results.

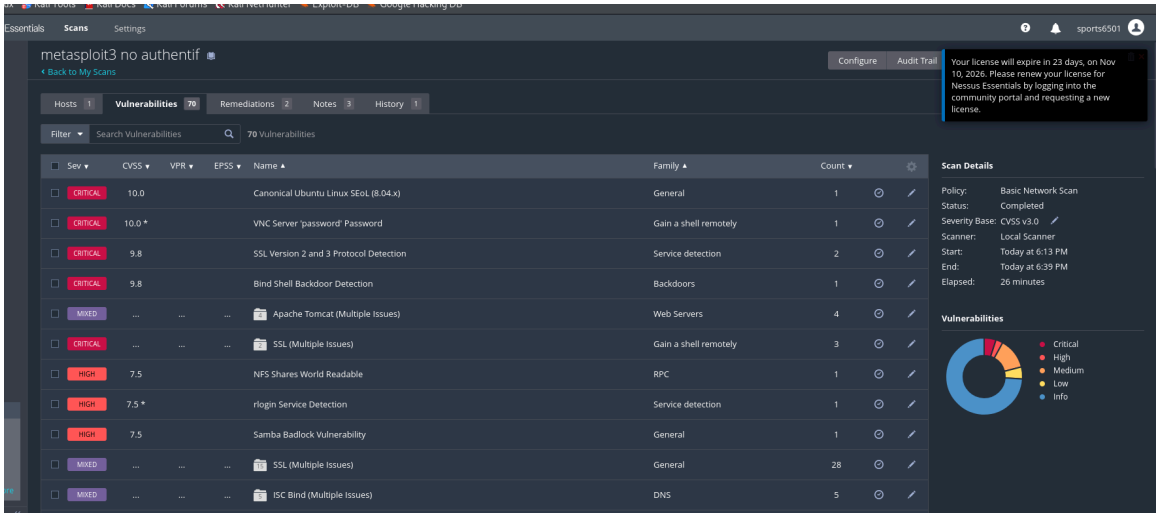
3.1 Environment & Scan Configuration

- **Nessus version:** Nessus Essentials
- **Common target:** Metasploitable 2 (192.168.56.101)
- **Scan A:** Basic Network Scan, unauthenticated (scan name: "metasploit3 no authentif")
- **Scan B:** Web Application Tests (scan name: "msf dvwa authent")
- **Port range (both scans):** default
- **Severity base:** CVSS v3.0 (both scans)

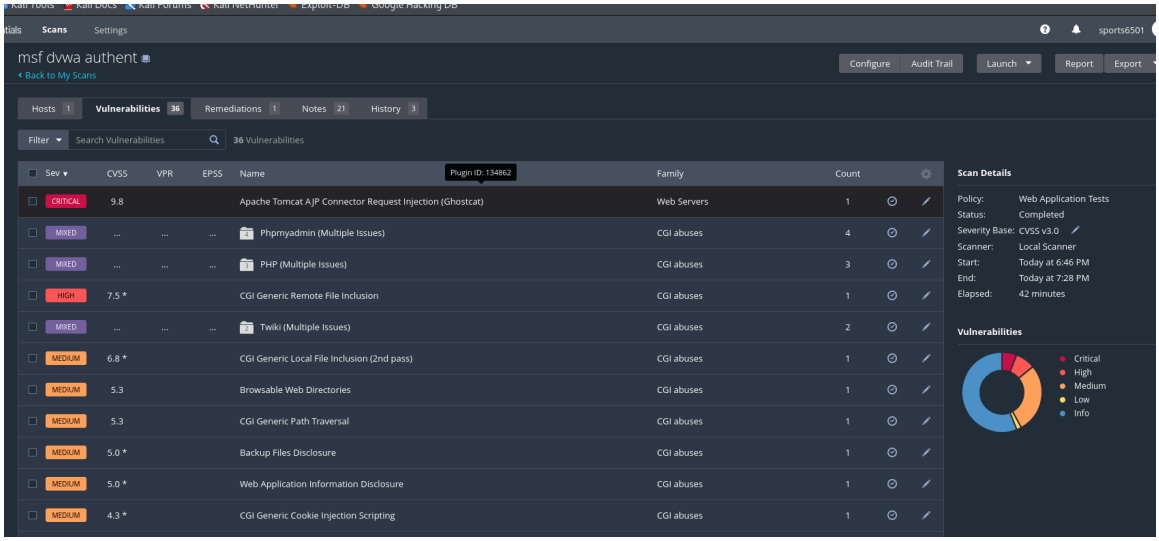
Screenshot: Basic Network Scan results

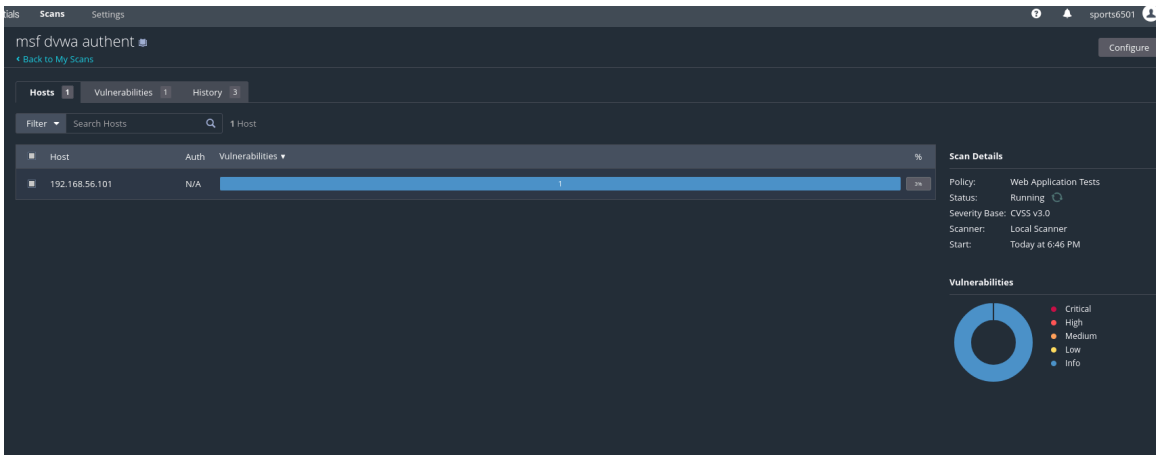


Screenshot: Web Application Tests results



Screenshots: scans in progress





3.2 Side-by-Side Results Comparison

Metric	Basic Network Scan	Web Application Tests
Scan duration	26 minutes (6:13 PM – 6:39 PM)	42 minutes (6:46 PM – 7:28 PM)
Total findings	70	36
Critical (visible in top list)	4 standalone plus grouped SSL/Tomcat entries	1
High (visible in top list)	3	1
Medium (visible in top list)	Mostly inside grouped "Mixed" entries	6
Low / Informational	94 info	21 info
CVSS v3 scores	10.0, 10.0, 9.8, 9.8, 7.5, 7.5, 7.5	9.8, 7.5, 6.8, 5.3, 5.3, 5.0, 5.0, 4.3
False positives	None noted	None noted

Note: exact per-severity totals use the exported reports for exact counts.

3.3 Analysis of Unique Findings

Template	Finding / CVE	Severity	Why the other scan did not report it
Basic Network Scan only	VNC Server 'password' Password	Critical (10.0)	Web application tests focus on HTTP/HTTPS and do not test the VNC service (port 5900) for default credentials.
Basic Network Scan only	Bind Shell Backdoor Detection	Critical (9.8)	Requires network-level service probing, which the web template does not emphasize.
Basic Network Scan only	NFS Shares World Readable	High (7.5)	Requires probing RPC/NFS services, outside the scope of a web crawler.
Web Application Tests only	CGI Generic Remote File Inclusion	High (7.5)	Requires crawling web directories, analyzing parameters and actively testing for file inclusion.
Web Application Tests only	Backup Files Disclosure	Medium (5.0)	Requires web crawling to discover exposed backup files in web-accessible directories.

Template	Finding / CVE	Severity	Why the other scan did not report it
Web Application Tests only	CGI Generic Path Traversal / Cookie Injection Scripting	Medium (5.3 / 4.3)	Generic CGI plugins are only enabled in the web template.

Verification needed: The earlier draft listed *Apache Tomcat AJP Connector Request Injection (Ghostcat), Critical 9.8* as unique to the web scan. In the Basic scan screenshot there is a grouped entry "Apache Tomcat (Multiple Issues)" with 4 findings, and Ghostcat may be inside that group. Expand the group before claiming Ghostcat was missed by the Basic scan.

3.4 Template Comparison

Criteria	Basic Network Scan	Web Application Tests	Notes
Scan duration	26 min	42 min	Web scanning takes longer due to crawling and testing parameters.
Types of vulnerabilities	Network / services	Web / HTTP / CGI	Basic: OS end-of-life, VNC, NFS, Samba, SSL, DNS, rlogin. Web: phpMyAdmin, PHP, Twiki, CGI abuses.
Plugin families emphasized	General, service detection, backdoors, RPC, DNS, SSL	CGI abuses, web servers	Basic scan does not run the intensive CGI/web plugins by default.
Credentials required	No	No (despite "authent" in the scan name)	Hosts view shows Auth = N/A for both.
Web analysis depth	Low	High	Web scan reports many CGI-abuse findings in the web applications.
Total findings	70	36	Basic covers more services; web goes deeper on HTTP.
False-positive rate	Not assessed	Not assessed	No findings were manually validated.

Conclusion

The two templates are complementary. The Basic Network Scan gives broad coverage of services, operating-system and configuration problems (end-of-life OS, VNC default password, bind shell backdoor, NFS), while the Web Application Tests template digs deeper into web applications (phpMyAdmin, PHP, Twiki, CGI weaknesses) at the cost of a longer scan. A thorough assessment should run both.

Deliverables Checklist

- ☒ Both scans executed against Metasploitable 2 (192.168.56.101)
- ☒ Results comparison table completed
- ☒ Unique findings analysis completed
- ☒ Template comparison completed
- ☐ Export both Nessus reports (PDF/HTML) using the Report/Export buttons Licence Nessus Essential no allow toexport reports